

Gimbal Thrust-Vector-Control Actuator Sizing for 316L Liquid Rocket Engines

Can two NEMA 17 steppers gimbal an LREKit-generated chamber and nozzle?
A torque, rate, and reflected-inertia study from 100 N to 100 kN, with a route into LREKit
and its MDO layer

Ibrahim Shahid

30 July 2026

Short answer. Two NEMA 17 steppers can gimbal an LREKit 316L chamber-and-nozzle assembly *only below roughly 3 kN of thrust*, and only with a correctly chosen reduction of about 15–25 (motor radians per gimbal radian). At 1 kN the design closes with a torque margin of 1.5; at 2 kN the margin is 1.18; by 5 kN there is *no* reduction ratio that satisfies torque and slew rate simultaneously. The binding constraint is almost never raw torque — gearing fixes torque for free — it is **shaft speed**, and behind it **reflected rotor inertia** and **mechanical power**, neither of which gearing can cheat. Independently of thrust, an open-loop stepper is the wrong device class for a flight control loop, because a lost step is an undetected control-surface position error.

Contents

1	The question, stated precisely	2
2	What a gimbal TVC actuator has to do	3
2.1	Deflection, rate, and bandwidth	3
2.2	The torque budget	3
2.3	A caveat on the moment arm	4
3	The engine side: mass properties from LREKit	4
3.1	Where the geometry comes from	4
3.2	The 316L shell	4
3.3	A consequence of L^* that dominates the inertia	5
3.4	Results	5
4	The actuator side	6
4.1	What a NEMA 17 actually delivers	6
4.2	Gearing buys torque and sells speed; power is conserved	7
4.3	Reflected inertia — the term that kills high reduction	7
4.4	The three screens	8

5	Verdict	8
5.1	Two NEMA 17s, directly answered	8
5.2	The map by thrust class	10
5.3	Why “yes, below 3 kN” is still the wrong thing to build	11
6	The 3D-printed demonstrator	11
6.1	Envelope	11
6.2	The 1 kN reference replica	12
6.3	The gimbal ring, and what the demo proves	12
7	Bringing TVC into LREKit: a raosim/tvc.py module	13
7.1	What already exists and what is missing	13
7.2	Proposed interface	13
7.3	CLI surface and output contract	14
8	The traditional solver and the MDO: what each one is	14
8.1	The traditional solver: a forward analysis pipeline	15
8.2	The MDO: a coupled, differentiable, inverse solve	15
8.3	Is the MDO part of the traditional solver?	16
8.4	What the MDO does that the traditional solver cannot	16
9	Adding TVC to the MDO	17
9.1	Why it belongs in the MDO at all	17
9.2	Three ways in, in increasing order of ambition	17
9.3	What has to be true for it to work	18
9.4	What I expect it to change	18
10	Limitations	18
11	Reproducing this	19

1 The question, stated precisely

The question “can I use two NEMA 17s for gimbal TVC” is under-determined as asked, because a motor and a gimbal are separated by a transmission whose ratio is a free design variable. Ask it the naive way — compare the motor’s 0.45 N m holding torque against the 4.3 N m the gimbal needs at 1 kN — and you conclude “no” by a factor of ten. That conclusion is wrong for the stated reason and right for a different one.

The question that actually has an answer is:

Does there exist a transmission ratio N (motor radians per gimbal radian) such that two NEMA 17 steppers simultaneously deliver the required gimbal torque, reach the required gimbal slew rate, and do not present the control loop with a reflected rotor inertia larger than the load they are supposed to be moving?

Those three requirements pull N in opposite directions, so they define a band $[N_{\min}, N_{\max}]$. If the band is non-empty the answer is yes; if it is empty, no amount of gearbox shopping helps. Section 4.4 formalises this. Sections 2 and 3 build the two sides of the inequality: what the vehicle demands, and what the engine’s mass properties are.

Everything on the engine side is generated by LREKit itself — the contours, the throat radii, the shell masses — so the study is reproducible for any design point the CLI can produce, not just the ones tabulated here.

2 What a gimbal TVC actuator has to do

2.1 Deflection, rate, and bandwidth

The design point used throughout this report is $\pm 5^\circ$ of gimbal deflection at a 4 Hz closed-loop bandwidth. Both numbers are taken from flight practice rather than invented:

- **Bandwidth 4 Hz.** The requirement envelope that NASA MSFC tested its electromechanical TVC actuator against was “4 Hz with 20–25 degrees of phase lag at 1 Hz”, stated in the source as *an SSME requirement* [1]. The same programme’s second-generation actuator carried a ≥ 4.2 Hz bandwidth requirement. A later MSFC electric-TVC integration test swept 0.5–16 Hz and reported good response to about 4 Hz on a deliberately under-powered system [2].
- **Deflection $\pm 5^\circ$.** Gimbal deflections in the 5–10° band are the conventional design range; the hobby-scale 3D-printed servo gimbal literature converges on $\pm 5^\circ$ as well [5].
- **Rate.** At $\pm 5^\circ$ and 4 Hz a sinusoid peaks at $\dot{\theta}_{\max} = \theta_{\max} \omega_c = 0.0873 \times 25.13 = 2.19$ rad/s = $126^\circ/\text{s}$, and $\ddot{\theta}_{\max} = \theta_{\max} \omega_c^2 = 55.1$ rad/s². For calibration: liquid landers are quoted around $20^\circ/\text{s}$ as a lower bound, missile nozzle actuators around $400^\circ/\text{s}$, and the hobby servo gimbal in [5] achieved $\pm 5^\circ$ in 44.5 ms, i.e. $\approx 225^\circ/\text{s}$. Our $126^\circ/\text{s}$ sits sensibly between the lander and hobby numbers.

2.2 The torque budget

Five terms act about the gimbal centre of rotation on each axis. In rough descending order of importance at this scale:

$$\tau_J = J_p \ddot{\theta} \quad \text{engine inertia about the pivot} \quad (1)$$

$$\tau_K = K_{\text{flex}} \theta \quad \text{propellant flex lines + gimbal flexure} \quad (2)$$

$$\tau_{\text{mis}} = F \ell \sin \delta \quad \text{thrust-line misalignment about the pivot} \quad (3)$$

$$\tau_{\text{unb}} = m a_{\text{veh}} d_{\text{cg}} \sin \theta \quad \text{engine mass unbalance under vehicle accel} \quad (4)$$

$$\tau_f = \mu F r_b \quad \text{gimbal bearing friction} \quad (5)$$

with J_p the transverse moment of inertia of the gimbale assembly about the pivot, ℓ the pivot-to-nozzle-exit distance, δ the thrust misalignment angle, d_{cg} the pivot-to-CG offset, and r_b the bearing journal radius. The sized actuator requirement is $\tau_{\text{sized}} = 2.0 \sum \tau_i$.

That the inertia and flex-line terms dominate — not thrust — is the standard result, and it is worth quoting the source verbatim because it is counter-intuitive to anyone who assumes the actuator “pushes against the thrust”:

“High powered TVC systems are needed to react against the inertia of an engine or nozzle, as well as against the stiffness of an engine gimbal or flex bearing, vehicle structure and propellant flex lines, at the slew rates necessary to maintain stable control of the vehicle.” — Bates & Young, NASA MSFC [2]

A well-designed gimbal puts the pivot on the thrust axis, so the thrust itself produces (to first order) no moment. What is left is inertia, spring rate, and friction.

Values used. K_{flex} is taken as 25% of the inertia term at full deflection. This is a placeholder and is flagged as such: on real engines the flex-line spring rate is *measured*, not predicted, because it depends on hose construction, coil geometry, and internal pressure. Bearing friction uses $\mu = 0.003$, a rolling-element value — the VINCI gimbal development chose needle roller bearings specifically “due to the low friction torque and the low wear of rolling bearings” [4]. A spherical plain bearing or rod-end would be $\mu \approx 0.05\text{--}0.15$, i.e. 20–50× worse, and at 100 kN that single substitution moves friction from a negligible term to a leading one. Thrust misalignment uses $\delta = 0.5^\circ$; mass unbalance uses $a_{\text{veh}} = 4g_0$. Wet assembly mass (injector, valves, manifolds, igniter, trapped propellant) is taken as $2.2\times$ the bare shell.

2.3 A caveat on the moment arm

The near-universal shortcut $\tau_g = F_a L_{\text{MA}}$ — actuator force times perpendicular moment arm — is exact only in a planar, one-degree-of-freedom sense. In a real two-axis gimbal, pitch motion tilts the yaw actuator’s plane of motion out of coplanarity, and part of the actuator force is then reacted by the gimbal joint rather than converted into useful torque. NASA MSFC’s TVC technical bulletin TB-03 derives the full 3D transformation and quantifies the error: discrepancies “become more pronounced” beyond $2\text{--}3^\circ$, and for the worked example the moment-arm method is off by about 2% at 15° of yaw [3]. At our $\pm 5^\circ$ design point the planar shortcut is acceptable; a design that later grows to $\pm 10^\circ$ should adopt the transformation-matrix form.

3 The engine side: mass properties from LREKit

3.1 Where the geometry comes from

Each thrust point is a real LREKit run. The invocation pattern is

```
PYTHONPATH=. python -m raosim --no-banner --backend numpy \
--propellant LOX/RP-1 --oxidizer LOX --fuel RP-1 \
--pc 3.0e6 --target-thrust <F> --epsilon 4.524 --pa-over-p0 0.033775 \
--length-pct 80 --l-star 1.0 --contraction-ratio 3.0 \
--minimum-cylindrical-length 0.03 --ru-factor 1.5 \
--material 316L --out builds/tvc/F<F>
```

which reuses the repository’s own literature-anchored parameter set: $P_c = 3.0$ MPa and sea-level matched expansion ($\varepsilon = 4.524$, $P_a/P_0 = 0.033775$) as in the 13 kN regression benchmark; $R_u = 1.5 R_t$ from the SP-8120 throat-approach radius-ratio bound and the standard Rao/Sutton bell convention; LOX/RP-1 chamber gas properties from Sutton & Biblarz RPE 9th ed. Table 5-5 via `raosim/propellants.py`; $L^* = 1.0$ m inside the SP-125 Table 4-1 range of 40–50 in for LOX/RP-1.

The post-processing lives in `scripts/tvc_sizing_study.py`, added by this study. It reads each run’s `contour.csv` and builds the shell.

3.2 The 316L shell

316L properties come from LREKit’s own materials catalogue (`raosim/materials.py`, entry **Stainless 316L**, sourced there to SP-125’s naming of stainless steel as a common thrust-chamber wall metal plus handbook properties): $\rho = 8000$ kg/m³, $S_y = 290$ MPa, $T_{\text{max}} = 1100$ K.

Wall thickness is set by thin-wall hoop stress with a hot-strength knockdown and a burst factor of safety, floored at a manufacturing minimum:

$$t = \max\left(t_{\min}, \frac{\text{FOS} \cdot P_c r_c}{k_{\text{hot}} S_y}\right), \quad \text{FOS} = 2.0, \quad k_{\text{hot}} = 0.50, \quad t_{\min} = 1.0 \text{ mm}. \quad (6)$$

The $k_{\text{hot}} = 0.5$ knockdown reflects that austenitic stainless retains roughly half its room-temperature yield in the 700–900 K band; this is a screening value, not a qualified allowable.

The shell is integrated as stacked annular disks along the contour. For each segment of axial extent dx between inner radius r_i and outer radius $r_o = r_i + t$:

$$dm = \rho \pi (r_o^2 - r_i^2) dx, \quad dI = dm \left[\frac{r_o^2 + r_i^2}{4} + \frac{dx^2}{12} + (x_c - x_p)^2 \right], \quad (7)$$

the bracketed terms being the annulus transverse inertia about its own diameter, the slab correction, and the parallel-axis shift to the pivot at x_p . The pivot is placed at the throat station throughout, which is close to the usual “gimbal block just downstream of the injector dome” practice and keeps J_p near its minimum.

3.3 A consequence of L^* that dominates the inertia

At fixed L^* and contraction ratio, chamber length $L_c \approx L^*/\text{CR}$ is **independent of thrust**. LREKit’s own MDO guide states this explicitly. At $L^* = 1.0$ m and $\text{CR} = 3$ that is a 0.33 m chamber whether the engine makes 100 N or 100 kN.

The practical effect is severe and easy to miss: at low thrust the engine is a long thin tube, its CG sits 130–165 mm *upstream* of the throat, and the parallel-axis term $m d_{\text{cg}}^2$ — not the shell’s own transverse inertia — sets J_p . This is why a 100 N engine still needs 1.2 N m of sized gimbal torque, which is more than a NEMA 17’s holding torque. Shortening the chamber (lower L^* , higher CR) is a far more effective lever on actuator size than lightening the wall. That is exactly the kind of cross-discipline trade the MDO layer exists to find, and it is the argument of Section 9.

3.4 Results

Table 1: 316L chamber-and-nozzle mass properties and the resulting per-axis gimbal torque budget. $\pm 5^\circ$ at 4 Hz, pivot at the throat, wet assembly factor 2.2, design margin 2.0.

F [kN]	t [mm]	R_t [mm]	D_e [mm]	L [mm]	m_{shell} [kg]	m_{gim} [kg]	J_p [kg m ²]	τ_{sized} [N m]	P_{peak} [W]
0.1	1.00	2.71	11.5	344	0.090	0.20	0.0072	1.23	2.7
0.25	1.00	4.29	18.3	351	0.139	0.31	0.0111	1.94	4.3
0.5	1.00	6.07	25.8	358	0.196	0.43	0.0156	2.83	6.2
1	1.00	8.58	36.5	368	0.280	0.62	0.0221	4.30	9.4
2	1.00	12.14	51.6	382	0.406	0.89	0.0318	6.98	15.3
5	1.38	19.19	81.6	411	0.936	2.06	0.0729	18.63	40.9
10	1.94	27.14	115.4	443	2.000	4.40	0.1568	44.18	96.9
15	2.38	33.23	141.4	467	3.146	6.92	0.2507	74.57	163.6
25	3.08	42.91	182.5	506	5.632	12.39	0.4690	147.46	323.4
50	4.35	60.68	258.1	578	12.693	27.93	1.2062	390.14	855.7
100	6.15	85.81	365.0	679	29.427	64.74	3.6352	1109.04	2432.4

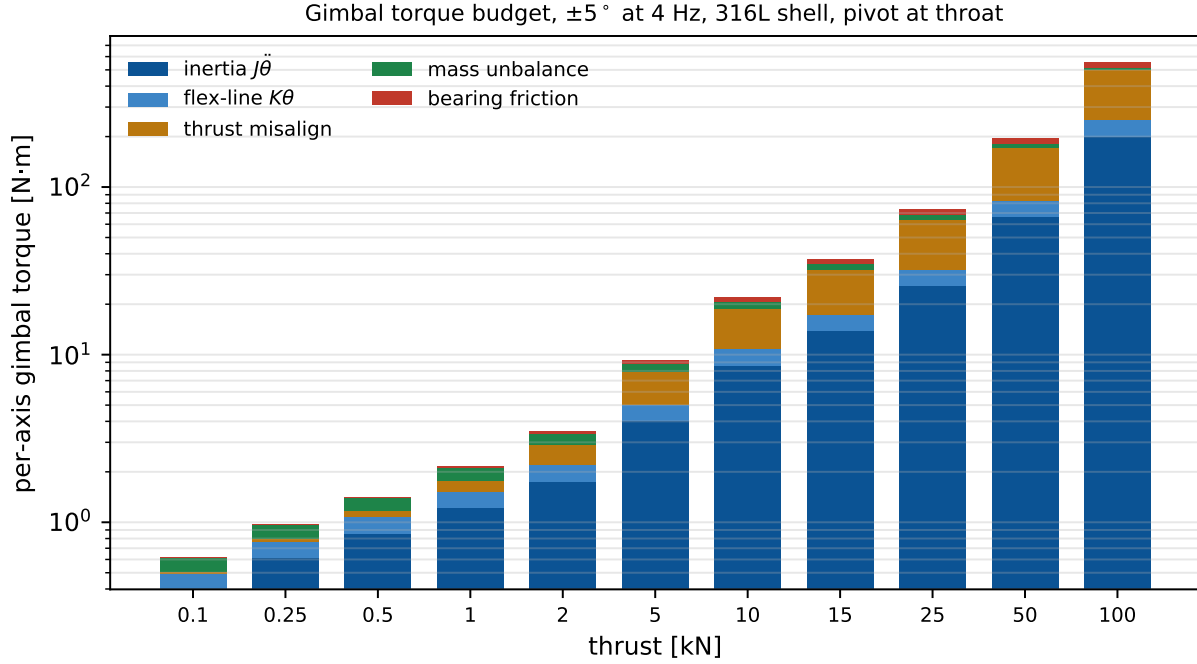


Figure 1: Torque budget by term. Inertia dominates below ~ 25 kN; above that the thrust-driven misalignment term overtakes it, because the pivot-to-exit moment arm grows with the engine while the rate requirement does not. At 100 kN the split is misalignment 252 N m, inertia 200 N m, flex line 50 N m, friction 41 N m, unbalance 11 N m.

For a sanity check against flight hardware: the VINCI upper-stage gimbal (a ~ 180 kN engine) used a friction-torque measurement rig with a 300 N m range and 10 N m accuracy [4]. Our 100 kN total sized requirement of 1109 N m against a friction component of 41 N m is the right order for a rolling-element gimbal at that scale.

4 The actuator side

4.1 What a NEMA 17 actually delivers

“NEMA 17” is a *frame size* (42×42 mm faceplate), not a performance class. Across the common catalogue the holding torque spans 16–79 N cm, i.e. 0.16–0.79 N m [7]. Three things about that number matter here:

1. **Holding torque is a standstill number.** It is what the motor resists at zero speed with rated current. The torque available while *moving* follows the pull-out curve and falls with step rate as winding inductance and back-EMF limit current rise. Datasheet pull-out curves for 42 mm frames typically show useful torque collapsing through the 600–1000 rpm region. This study uses 0.20 N m at 600 rpm for a 0.45 N m motor and 0.35 N m at 600 rpm for a 0.79 N m motor — roughly 45% of holding torque, which is a conventional and not especially pessimistic derate.
2. **Rotor inertia is not negligible.** A 42 mm frame carries $J_{\text{rotor}} \approx 5.4 \times 10^{-6}$ to 1.4×10^{-5} kg m². Section 4.3 shows why this is the term that ends the argument.
3. **It is open loop.** The controller assumes the motor is where it was told to go. When torque is momentarily inadequate the motor skips steps and the controller does not know. Industrial

guidance is explicit that this disqualifies open-loop steppers from safety-critical positioning: “should the machine snag on an object, [a servo] will be sensed immediately... and never lose position”, whereas an open-loop stepper “does not know the actual location of the machine” [6]. A TVC actuator that silently loses 3° of commanded gimbal is a loss-of-vehicle event.

4.2 Gearing buys torque and sells speed; power is conserved

The transmission ratio N multiplies torque and divides speed. Mechanical power passes through essentially unchanged (times efficiency η):

$$\tau_{\text{gimbal}} = N\eta\tau_{\text{motor}}, \quad \omega_{\text{motor}} = N\omega_{\text{gimbal}}, \quad P = \tau\omega \text{ invariant.} \quad (8)$$

For a screw-and-lever transmission with lead L and moment arm R , $N = 2\pi R/L$. A 60 mm arm on a 2 mm-lead screw gives $N = 188$; on an 8 mm lead, $N = 47$.

This is why the power column of the results table is the most informative single number in this report. Two NEMA 17s at 0.20 N m and 600 rpm produce $2 \times 0.20 \times 62.8 = 25$ W of shaft power. The requirement is 9.4 W at 1 kN, 41 W at 5 kN, 97 W at 10 kN. No gearbox changes those numbers.

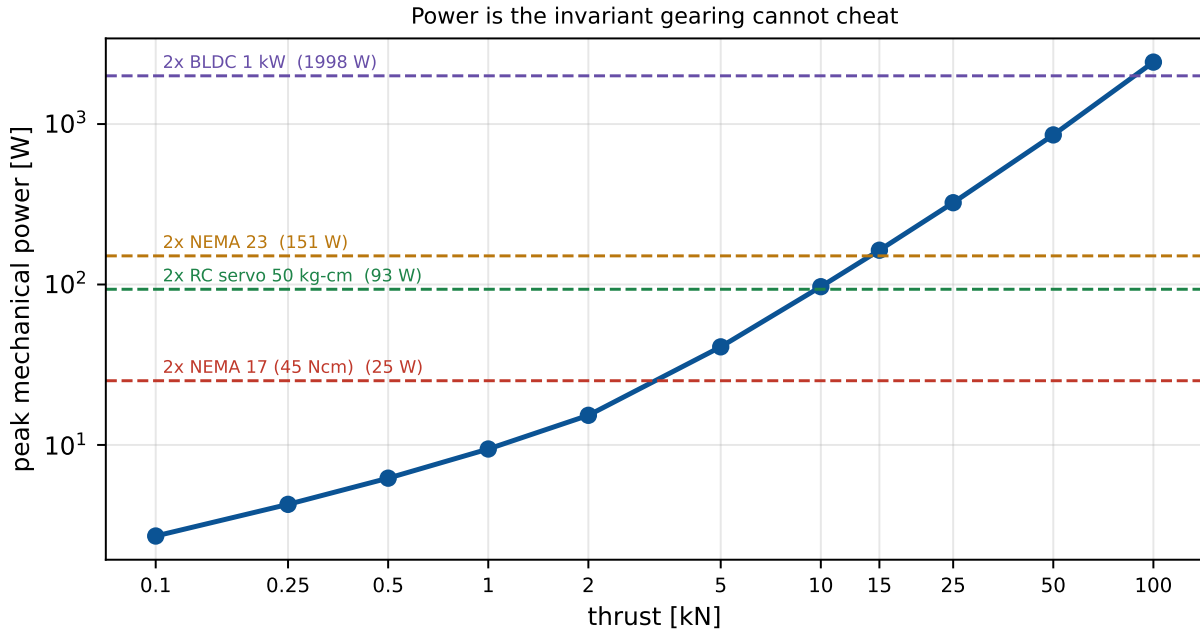


Figure 2: Required peak mechanical power per axis versus available shaft power for four candidate actuator pairs. Crossings here are hard limits; crossings on torque alone are not.

4.3 Reflected inertia — the term that kills high reduction

Referred to the gimbal axis, the motor’s own rotor contributes $N^2 J_{\text{rotor}}$ of apparent inertia. Because it scales as N^2 while torque scales as N , past a certain ratio the motor spends more torque accelerating itself than the engine. NASA MSFC found precisely this while designing their second-generation EMTVC actuator:

“Since inertia is a major concern due to the EMA’s nature of operation, optimization of horsepower, RPM, and motor rotor diameter is imperative. . . . Calculations show that the amount of inertia created by the rest of the system is essentially negligible when compared to the inertia created by the cyclic action of each rotor inertia.” — Cowan & Weir, NASA MSFC [1]

Motion-control practice caps the load-to-rotor inertia ratio by device class [6]: about 10× for an open-loop stepper, 30× for a closed-loop stepper, and 100× for a true servo. Inverting gives the ceiling

$$N_{\max}^{(J)} = \sqrt{\frac{k J_p}{J_{\text{rotor}}}}, \quad k \in \{10, 30, 100\}. \quad (9)$$

Worked at 1 kN with two 45 N cm NEMA 17s ($J_{\text{rotor}} = 1.08 \times 10^{-5} \text{ kg m}^2$ for the pair, $J_p = 0.0221 \text{ kg m}^2$): $N_{\max}^{(J)} = 143$. Take instead the 60 mm-arm / 2 mm-lead screw with $N = 188$ and the reflected rotor inertia is $188^2 \times 1.08 \times 10^{-5} = 0.38 \text{ kg m}^2$ — seventeen times the engine it is supposed to be moving.

4.4 The three screens

Collecting the three requirements as bounds on N :

$$\text{torque: } N \geq N_{\min}^{(\tau)} = \frac{\tau_{\text{sized}}}{\eta \tau_{\text{motor}}} \quad (10)$$

$$\text{speed: } N \leq N_{\max}^{(\omega)} = \frac{\omega_{\text{motor,max}}}{\omega_{\text{gimbal}}} \quad (11)$$

$$\text{inertia: } N \leq N_{\max}^{(J)} = \sqrt{k J_p / J_{\text{rotor}}} \quad (12)$$

The actuator is feasible iff $N_{\min}^{(\tau)} \leq \min(N_{\max}^{(\omega)}, N_{\max}^{(J)})$. This is implemented in `screen_actuator()` in `scripts/tvc_sizing_study.py`, with $\eta = 0.85$.

5 Verdict

5.1 Two NEMA 17s, directly answered

Table 2: Feasible band in N for two NEMA 17 (45 N cm) steppers. N^* is the geometric mid-band pick; “margin” is delivered torque over required torque at N^* .

F [kN]	$N_{\min}^{(\tau)}$	$N_{\max}^{(\omega)}$	$N_{\max}^{(J)}$	N^*	verdict
0.1	3.6	28.6	81.7	10.2	yes comfortable (margin 2.8)
0.5	8.3	28.6	120.2	15.4	yes (margin 1.86)
1	12.6	28.6	143.2	19.0	yes tight (margin 1.50)
2	20.5	28.6	171.6	24.2	yes marginal (margin 1.18)
5	54.8	28.6	259.8	—	no band empty — speed-limited
10	129.9	28.6	381.0	—	no
25	433.7	28.6	659.0	—	no
100	3261.9	28.6	1834.6	—	no

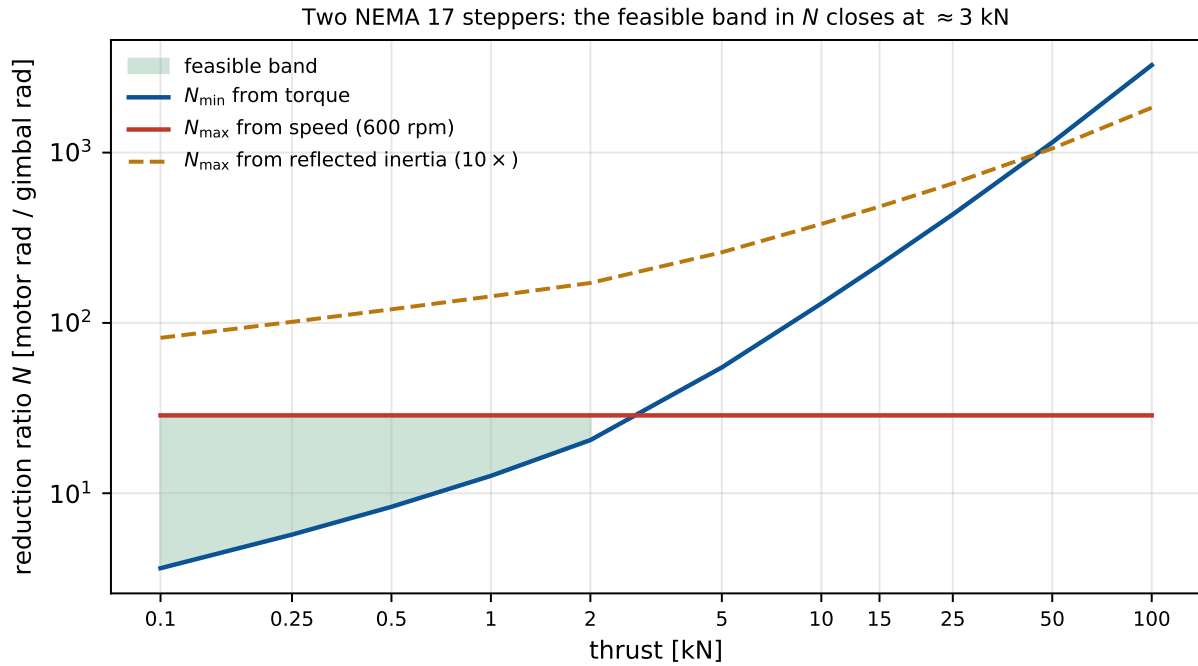


Figure 3: The feasible band in reduction ratio for two NEMA 17s. The torque floor rises with thrust; the speed ceiling is flat because the gimbal rate requirement does not depend on engine size. They cross near 3 kN.

Note what the crossing is *not*: it is not the reflected-inertia limit (the dashed line stays an order of magnitude above the torque floor throughout the feasible region, and only meets it near 50 kN where the design has been infeasible for a decade of thrust already), and it is not raw torque. It is that the stepper runs out of shaft speed. That is the physically interesting result, and it is invisible if you only compare holding torque against required torque.

5.2 The map by thrust class

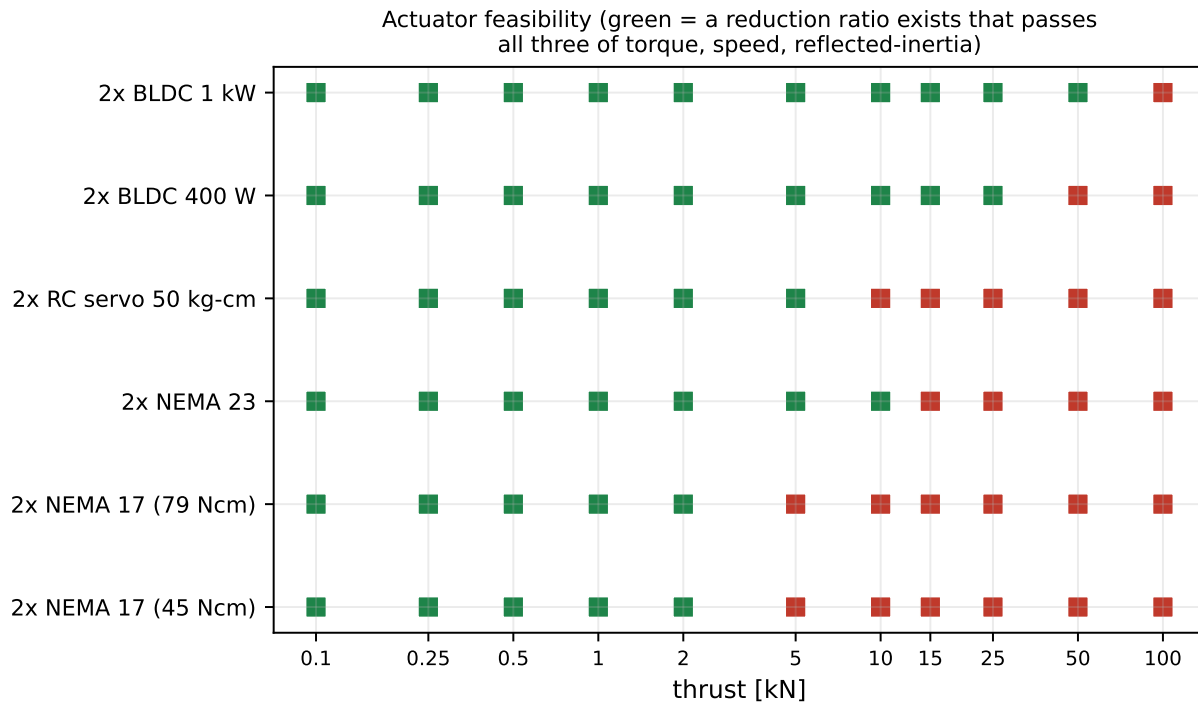


Figure 4: Feasibility of six candidate actuator pairs across the sweep.

Table 3: Recommended actuation architecture by thrust class.

Class	Actuator	Why, and what to watch
100–500 N small	2× high-torque digital RC servo (25–50 kg cm, metal gear), direct crank $N \approx 1$ –2	$\tau_{\text{sized}} \leq 2.8 \text{ N m}$, $P \leq 6 \text{ W}$. Closed-loop by construction, $\sim \$45$ each, and no gearbox to design. NEMA 17s also close here but buy you nothing. Watch: servo gear backlash appears directly as gimbal deadband.
1–10 kN medium	2× brushless RC servo (50 kg cm) to $\sim 7 \text{ kN}$; 2× NEMA 23 closed-loop, $N \approx 20$ –35, to 10 kN; BLDC + planetary above that	$\tau_{\text{sized}} = 4.3 \rightarrow 44 \text{ N m}$, $P = 9 \rightarrow 97 \text{ W}$. This is where the choice actually matters. The NEMA 23 solution at 10 kN closes with margin 1.15 — too thin; prefer a 400 W BLDC with a 20:1 planetary (margin > 5 , $P_{\text{req}}/P_{\text{avail}} = 0.14$). Watch: at these powers the flex-line spring rate must be <i>measured</i> , not assumed.
>15 kN large	BLDC servo + roller screw, i.e. the MSFC EMTVC architecture in miniature; above $\sim 50 \text{ kN}$, EMA/EHA/hydraulic	$\tau_{\text{sized}} = 75 \rightarrow 1109 \text{ N m}$, $P = 164 \rightarrow 2432 \text{ W}$. At 100 kN even two 1 kW BLDC servos fail ($P_{\text{req}}/P_{\text{avail}} = 1.43$). Roller screws are specifically preferred over ball screws here for start-transient shock capability [1]. Watch: this class needs a real failure analysis — ball/roller screw seizure is a single-point-of-failure mode [2].

5.3 Why “yes, below 3 kN” is still the wrong thing to build

Even inside the feasible band, an open-loop stepper is a poor TVC actuator:

- **Silent failure mode.** A skipped step is an unmeasured position error, and the flight computer’s estimate of the thrust vector is now wrong in a way nothing detects [6].
- **Torque margin is worst exactly when it matters.** Available torque falls as speed rises; a large disturbance demands both simultaneously.
- **Resonance.** Open-loop steppers have mid-band instability regions; a lightly damped gimbal is a good place to find one.
- **Holding current.** A stepper holds position by dissipating full rated current at zero output power — a poor thermal and battery choice for a device that spends most of a burn near null.

If steppers are chosen anyway for cost or availability, add an encoder and close the loop; that lifts the inertia ratio limit from $10\times$ to $30\times$, raises usable speed, and removes the silent-failure mode.

6 The 3D-printed demonstrator

6.1 Envelope

Table 4: Print envelope against a Bambu Lab P1S (256³ mm, 240 mm usable). Diameters include the 316L-equivalent wall.

F [kN]	L [mm]	D_{exit}	D_{cham}	parts	note
0.1	344	13.5	11.4	2	split at flange
0.5	358	27.8	23.0	2	split at flange
1	368	38.5	31.7	2	reference replica
5	411	84.4	69.2	2	split at flange
10	443	119.3	97.9	2	split at flange
15	467	146.1	119.9	2	split at flange
25	506	188.7	154.8	3	split at flange
50	578	266.8	218.9	3	exceeds bed diameter; print at $0.90\times$
100	679	377.3	309.6	3	exceeds bed diameter; print at $0.64\times$

Every case is length-limited, and for the same reason as Section 3: the L^* -driven chamber is thrust-independent. Diagonal placement technically fits parts up to 443 mm but needs full-height support against the one surface the demonstrator exists to show, so it is not recommended.

6.2 The 1 kN reference replica

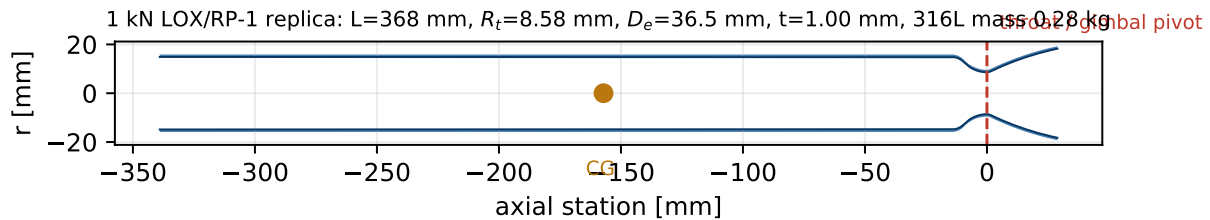


Figure 5: 1 kN LOX/RP-1 sea-level-matched contour with the hoop-sized wall, throat/pivot station, and CG. The CG sits 157 mm upstream of the pivot — that offset, not the shell’s own inertia, is what the actuator fights.

Throat radius	8.58 mm
Exit diameter	36.5 mm (outer 38.5 mm)
Chamber diameter	29.7 mm (outer 31.7 mm)
Overall length	368 mm
Wall thickness	1.00 mm (hoop floor)
316L mass (if metal)	0.280 kg
PLA mass (printed, $\rho = 1240$)	0.043 kg
Split	2 parts at the chamber/converging flange, ≈ 184 mm each
Print settings	0.2 mm layer, 3 walls, 15% gyroid, no support with the nozzle printed exit-down (all overhangs $< 45^\circ$ on the bell)
Estimated time	$\sim 6\text{--}8$ h per half at 0.2 mm on a P1S

Print it in **PETG or ASA**, not PLA. The arXiv hobby-gimbal study moved from PLA to ABS for exactly this reason — PLA’s low glass transition made sustained operation near the motor a concern, and ABS is both less dense (1.04 vs 1.24 g/cm³) and tougher [5]. The P1S is enclosed, so ASA/ABS is a reasonable choice on this printer specifically.

6.3 The gimbal ring, and what the demo proves

Build the two-axis ring around the throat station, not the chamber: that puts the pivot where the analysis puts it and keeps J_p at the value the torque budget assumed. Two orthogonal pivot axes, actuator horns at a 50–70 mm arm, and hard stops at $\pm 6^\circ$.

Be precise about what the demonstrator does and does not show. A PLA/PETG replica has $\sim 15\%$ of the steel mass and therefore $\sim 15\%$ of the inertia. It *does* validate: kinematics, actuator travel and interference, control-loop tuning topology, sensor integration, mechanism backlash, and the gimbal-ring geometry. It *does not* validate any torque, rate, or power number in this report. To make the bench test quantitative, ballast the printed part to the 316L wet mass *and* match J_p — matching mass alone is not sufficient, because the parallel-axis term dominates and depends on where you put the ballast.

7 Bringing TVC into LREKit: a raosim/tvc.py module

The analysis above is currently a script that post-processes `contour.csv`. Everything it needs already exists inside LREKit; the natural next step is to make it a first-class module so that any design point the CLI produces carries a TVC verdict.

7.1 What already exists and what is missing

Needed	Status	Where
Wall contour (x, r)	present	<code>design_nozzle_v2</code> $\rightarrow$ <code>contour.csv</code>
Wall thickness, station-wise	present	<code>thermal_design.py</code> , <code>regen_profile.py</code>
Material density and S_y	present	<code>materials.py</code>
Enclosed solid volume	present	<code>export.py</code> STL diagnostics
Injector/valve/manifold mass	<i>partly</i>	<code>injector_cad.py</code> has geometry, not mass
Pump/battery mass	present	<code>pumps.py</code> , <code>mdo/pump.py</code>
Mass properties (CG, J)	missing	— this module
Gimbal torque budget	missing	— this module
Actuator screen	missing	— this module
Flex-line spring rate	missing, and not predictable	must be a user input or a measured table

7.2 Proposed interface

```
# raosim/tvc.py

@dataclass(frozen=True)
class MassProperties:
    mass_kg: float
    x_cg_m: float
    I_transverse_cg: float
    I_transverse_pivot: float
    pivot_x_m: float
    components: dict[str, float]  # shell / jacket / injector / trapped prop

def mass_properties(contour, thickness, material, *,
                   pivot="throat", extras=()) -> MassProperties: ...

@dataclass(frozen=True)
class GimbalRequirement:
    theta_max_deg: float = 5.0
```

```

bandwidth_hz: float = 4.0          # SSME envelope, NTRS 19940025147
k_flex_nm_per_rad: float | None = None # None -> reported as UNPINNED
mu_bearing: float = 0.003
misalignment_deg: float = 0.5
vehicle_accel_g: float = 4.0
design_margin: float = 2.0

def torque_budget(mp, thrust_n, req) -> TorqueBudget: ...
def screen_actuators(budget, catalog=ACTUATORS) -> list[Screen]: ...

```

7.3 CLI surface and output contract

```

--tvc                      enable the TVC block
--tvc-deflection 5.0        [deg]
--tvc-bandwidth 4.0        [Hz]
--tvc-pivot throat|injector|cg|<x in m>
--tvc-flex-stiffness <N.m/rad> measured; omit to report the term unpinned
--tvc-actuator-catalog <json> override the built-in candidate list
--tvc-dry-mass-extras <json> injector/valve/manifold masses

```

and a summary.json block:

```

"tvc": {
  "mass_properties": {"mass_kg":..., "x_cg_m":..., "I_pivot_kgm2":...,
    "pivot_x_m":..., "components":{...}},
  "requirement": {"theta_max_deg":5.0, "bandwidth_hz":4.0, ...},
  "torque_budget": {"inertia":..., "flex":..., "friction":...,
    "misalignment":..., "unbalance":...,
    "total":..., "sized":..., "peak_power_w":...},
  "actuators": [{"name":..., "feasible":true, "N_band":[lo,hi],
    "binding":"speed", "margin":1.50}, ...],
  "gates": {"flex_stiffness_pinned": false,
    "moment_arm_model_valid": true,
    "any_actuator_feasible": true}
}

```

Release gates. The repository’s convention is that a physically unqualified number must be gated, not quietly reported. Three gates belong here:

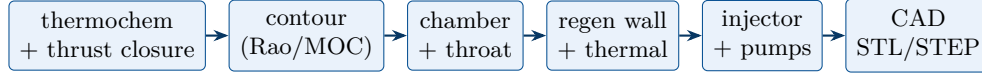
1. `flex_stiffness_pinned` — FAIL unless `--tvc-flex-stiffness` was supplied from measurement. The 25%-of-inertia placeholder used in this report is a screening assumption and must never pass as literature.
2. `moment_arm_model_valid` — WARN above $\theta_{\max} > 5^\circ$, where the planar moment-arm relation starts to degrade [3].
3. `actuator_open_loop` — WARN whenever the selected candidate is an open-loop stepper, regardless of margin.

8 The traditional solver and the MDO: what each one is

This section answers the three questions directly: what the traditional solver does, what the MDO does that it cannot, and whether the MDO is supposed to be part of the traditional solver.

8.1 The traditional solver: a forward analysis pipeline

`raosim/run_nozzle.py` (roughly 3300 lines, reached by `python-mraosim`) is a **sequential forward analysis**. You give it a complete specification — P_c , ε , target thrust, L^* , contraction ratio, channel geometry, pintle diameter, coolant, materials — and it walks a fixed chain:



Its properties:

- **One direction.** Information flows left to right. If the wall sizing comes back infeasible, nothing upstream changes — the run reports the failure and stops. *You* are the optimiser.
- **Highest-fidelity models.** This is where the exact variational Rao solve, method-of-characteristics, station-wise thermostructural wall sizing, machined-pintle CAD, and pump B-reps live.
- **It is where artifacts come from.** STL, STEP, contour CSV, manufacturing reports. The MDO layer produces none of these.
- **Local coupling only.** A few loops are closed internally (the spray $\rightarrow c^*$ fixed point, the channel-height sweep), but the disciplines are not solved simultaneously.

8.2 The MDO: a coupled, differentiable, inverse solve

`raosim/mdo/` is a different *kind* of object. Its own package docstring calls it “a pure-numerical layer: no CAD, no file I/O in the differentiated path”. It does two things the forward pipeline structurally cannot:

(1) **It closes the couplings simultaneously (MDF).** `engine.py:solve_engine` evaluates every block — nozzle, cooling, injector, pump/battery — as one coupled residual $R(y, x) = 0$ and solves it with Newton, so the cooling Δp that the pump must supply and the pump discharge pressure that the cooling channels see are consistent at the solution rather than after a couple of hand iterations. Sensitivities come from the *implicit function theorem*, never from unrolling the solver:

$$\frac{\partial y}{\partial x} = - \left(\frac{\partial R}{\partial y} \right)^{-1} \frac{\partial R}{\partial x}. \quad (13)$$

(2) **It inverts the question.** The forward solver answers “given this design, is it feasible?”. `nlp.py` answers “*find* the design that minimises feed-system mass subject to $I_{sp} \geq \text{floor}$ and all 17 constraints”. It is an ε -constraint NLP over a 10-variable unit box with exact JAX Jacobians handed to SLSQP, and it can trace a Pareto frontier rather than returning one point.

The 10 design variables are P_c , ε , the two injector Δp fractions, pintle diameter, pump rpm, channel width and height, film-cooling fraction, and wall thickness. The 17 constraints span separation, coking, channel fit, chug, pintle transition branch, pump suction and tip speed, aspect ratio, blockage band, structural stress, wall temperature, film capacity, property-table domain, Rao-chart domain, and wall monotonicity.

8.3 Is the MDO part of the traditional solver?

No — and it is not a replacement either. They are two independent front ends over a shared physics vocabulary, joined by an explicit bridge. The MDO *searches*; the traditional solver *resolves and builds*. The guide states the relationship plainly: “The traditional nozzle/CAD workflow is still the default.”

Concretely, the two live side by side on the same CLI:

```
lrekit                                # traditional forward workflow (default)
lrekit --engine-mdo                   --pc 3e6 ... # one coupled MDO point evaluation
lrekit --engine-mdo-optimize --isp-min 200 # the constrained optimisation
```

and are connected by two contracts:

- `mdo/snapshot.py` — `EngineAnalysisSnapshot`, a versioned description of an analysed engine that *both* sides can emit (`snapshot_from_mdo`, `snapshot_from_traditional`) and that `compare_snapshots` can diff.
- `mdo/postprocess.py` — the Phase-11 bridge. It maps an optimised `EngineState` back into a `design_nozzle_v2` input, pins the exact chamber thermochemistry so the two sides cannot silently drift apart, re-evaluates at higher fidelity, and emits a **parity report**.

The reason the bridge exists is that the two sides deliberately do not use the same nozzle model. The MDO’s `grid.py` uses the analytic Rao/TOP approximation — fast, differentiable, piecewise- C^1 — because a full variational Rao solve inside an SLSQP inner loop would be intractable. The authoritative post-optimum contour then comes from `design_nozzle_v2`. The parity report is what keeps that substitution honest.

8.4 What the MDO does that the traditional solver cannot

	Traditional solver	MDO layer
Question answered	“is this design feasible, and what does it look like?”	“what is the best feasible design?”
Direction	forward analysis	inverse / optimisation
Coupling	sequential, a few local fixed points	simultaneous (MDF), Newton + IFT
Derivatives	none	exact, by JAX AD through the implicit solve
Constraints	checked and reported after the fact	<i>enforced</i> by the NLP, all 17
Trade studies	one run per point, by hand	Pareto frontier in one call
Fidelity	highest (exact Rao, MOC, station-wise wall)	reduced, differentiable surrogates
Artifacts	STL, STEP, CSV, reports	none — numbers only
Backend	NumPy or JAX	JAX only

The distinction that matters most: **gradients**. Without them, exploring a 10-dimensional design space against 17 constraints means finite differences — at minimum 11 forward solves per gradient, each of which is a full Newton-converged coupled engine. With the implicit function theorem it is one solve plus one linear system. That is what turns “run a parameter sweep overnight” into “optimise in a few minutes”, and it is the whole reason the MDO layer is written in JAX rather than being another mode of `run_nozzle.py`.

The second distinction: **the MDO can find couplings you would not think to check by hand.** The film-cooling fraction is simultaneously a coking lever, an I_{sp} penalty, a coolant-flow change, and therefore a pump-power and battery-mass change. A human sweeping one variable at a time will not land on the joint optimum. A gradient will.

9 Adding TVC to the MDO

Once `raosim/tvc.py` exists on the traditional side, moving it into the MDO is the interesting part — because TVC is exactly the kind of constraint that couples to variables you would not expect.

9.1 Why it belongs in the MDO at all

The actuator requirement is driven by J_p , which is driven by chamber length and wall thickness, which are driven by L^* , contraction ratio, P_c , and the thermostructural wall sizing. Every one of those is either already an MDO design variable or one step away from it. Section 3 showed the specific consequence: at low thrust the L^* -driven chamber length, not the wall mass, sets the actuator size. A designer optimising for I_{sp} and feed mass alone will happily choose a long chamber and hand the TVC engineer a bill they never see.

9.2 Three ways in, in increasing order of ambition

(a) As a reported output only. Add J_p , τ_{sized} , and P_{peak} to `EngineState` and report them. Zero risk, no new design variables, and immediately useful — you can read the TVC cost of every point on an existing Pareto frontier.

(b) As a hard constraint. Add

$$g_{\text{tvc}}(x) = 1 - \frac{\tau_{\text{sized}}(x)}{\tau_{\text{actuator}}} \geq 0 \quad (14)$$

to `CONSTRAINT_NAMES` and `DEFAULT_ENFORCED`, with τ_{actuator} a mission input. This says “design me the best engine that my chosen actuator can actually gimbal”. It is the formulation that matches how a real vehicle programme works, because the actuator is usually selected before the engine is frozen.

(c) As a mass term in the objective. The existing objective is smooth electric-feed mass. Adding actuator and gimbal-structure mass — both of which scale with τ_{sized} — makes the optimiser trade chamber length against actuator mass on its own. This is the most valuable and the most delicate, because it needs an actuator mass-versus-torque scaling law that is honest. That law should come from a fitted catalogue with its fit residual reported, not from a guess.

9.3 What has to be true for it to work

Requirement	How to satisfy it
Differentiable, no branches	The mass integral is a smooth quadrature over the station grid — fine. The <i>screen</i> is not: <code>min</code> , <code>max</code> , and the feasibility predicate are non-smooth. Put the screen in post-processing and expose only the smooth $\tau_{\text{sized}}(x)$ to the NLP.
Fixed shapes, no Python control flow	<code>mdo/grid.py</code> already gives a fixed-topology station grid; integrate on it directly rather than on the variable-length <code>contour.csv</code> .
No zero Jacobian columns	The guide’s own dead-variable check (§8) must be re-run after adding the constraint. If $\partial g_{\text{tvc}}/\partial x_k = 0$ for every k , the constraint is decoration.
A NumPy oracle + parity test	Repository rule: every JAX block keeps a NumPy oracle. <code>scripts/tvc_sizing_study.py</code> as written <i>is</i> that oracle — it is pure Python/NumPy. Add <code>tests/test_mdo_tvc.py</code> asserting parity to 10^{-8} .
Snapshot round-trip	Add the TVC block to <code>EngineAnalysisSnapshot</code> so <code>compare_snapshots</code> covers it and MDO/traditional cannot drift.

9.4 What I expect it to change

A concrete, falsifiable prediction: with (b) or (c) enabled at the 1 kN scale, the optimiser should push *down* on chamber length — higher contraction ratio at fixed L^* , or a lower L^* where combustion efficiency permits — because J_p is dominated by the $m d_{\text{cg}}^2$ parallel-axis term. That would trade a small c^* efficiency loss against a large actuator saving, and it is precisely the kind of trade the forward solver cannot find because L^* and actuator torque never meet in the same calculation. If the optimiser does *not* do this, the constraint is probably not wired to the variables it should be, and the dead-column check above will say so.

10 Limitations

Stated plainly, because a portfolio piece that overclaims is worse than one that is narrow and honest:

1. **The flex-line spring rate is a placeholder.** 25% of the inertia term is a screening assumption with no source. On real engines this term is measured. It could plausibly be $2\times$ larger, which would move the NEMA 17 crossover from ~ 3 kN to ~ 2 kN. This is the single largest uncertainty in the report.
2. **Stepper pull-out torque is a derate, not a measurement.** 45% of holding torque at 600 rpm is conventional but motor-, driver-, and bus-voltage-specific. A 48 V driver on a low-inductance winding does considerably better than a 12 V one.
3. **Only the shell is integrated.** Injector, valves, manifolds and trapped propellant enter through a single $2.2\times$ factor. A real design would integrate them individually — and their mass sits near the injector end, i.e. at large d_{cg} , so this factor is doing more work than its simplicity suggests.
4. **Contours are seed geometry.** Runs used `-max-nfev 0`, so the contours are the analytic Rao/TOP seed, not the converged variational solve. Mass properties are insensitive to that at the percent level; performance numbers would not be.

5. **No structural dynamics.** The gimbal ring, actuator mounts, and vehicle backstructure have modes. The published rule of thumb is to keep the closed-loop bandwidth below one tenth of the backstructure mode frequency; the MSFC test article showed load-structure resonance at 8–9 Hz against a 4 Hz bandwidth [1]. Nothing here models that.
6. **No thermal environment on the actuator.** Radiation from a 316L nozzle at 1000 K is not modelled and is a real constraint on where the actuators can be mounted.

11 Reproducing this

```
# 1. generate contours (one per thrust point)
for F in 100 250 500 1000 2000 5000 10000 15000 25000 50000 100000; do
  PYTHONPATH=. python -m raosim --no-banner --backend numpy \
    --propellant LOX/RP-1 --oxidizer LOX --fuel RP-1 \
    --pc 3.0e6 --target-thrust $F --epsilon 4.524 --pa-over-p0 0.033775 \
    --length-pct 80 --l-star 1.0 --contraction-ratio 3.0 \
    --minimum-cylindrical-length 0.03 --ru-factor 1.5 \
    --max-nfev 0 --n-control 8 --n-kernel 12 \
    --material 316l --out builds/tvc/F$F
done

# 2. run the sizing study
python scripts/tvc_sizing_study.py --contours builds/tvc \
  --json builds/tvc/study.json --theta 5 --bandwidth 4
```

References

- [1] J. R. Cowan and R. A. Weir, “Design and Test of Electromechanical Actuators for Thrust Vector Control,” NASA George C. Marshall Space Flight Center, NTRS 19940025147. <https://ntrs.nasa.gov/api/citations/19940025147/downloads/19940025147.pdf> (*SSME 4 Hz / 20–25° bandwidth requirement; rotor-inertia dominance; roller-screw selection.*)
- [2] L. B. Bates and D. T. Young, “Developmental Testing of Electric Thrust Vector Control Systems for Manned Launch Vehicle Applications,” *Proc. 41st Aerospace Mechanisms Symposium*, JPL, May 2012. NASA MSFC, NTRS 20120014541. <https://ntrs.nasa.gov/api/citations/20120014541/downloads/20120014541.pdf> (*TVC reacts against inertia, gimbal/flex-bearing stiffness, structure and propellant flex lines; EMA/EHA/IAP architectures; ball-screw single-point-of-failure.*)
- [3] N. A. Stepp, “MSFC-TVC | TB-03: Derivation of Thrust Vector Control (TVC) Actuator-Force / Gimbal-Torque Transformation Matrix,” NASA/TM-20240012525, Marshall Space Flight Center, December 2024. <https://ntrs.nasa.gov/citations/20240012525> (*Planar moment-arm relation degrades beyond 2–3°; ≈2% error at 15° yaw in the worked example.*)
- [4] C. Neugebauer, M. Falkner, L. Supper and G. Traxler, “Bearing Development for a Rocket Engine Gimbal,” *Proc. 38th Aerospace Mechanisms Symposium*, ESMATS, 2006. <https://esматы.eu/amspapers/pastpapers/pdfs/2006/neugebauer.pdf> (*VINCI upper-stage gimbal; rolling bearings chosen for low friction torque; 300 Nm friction measurement range, 10 Nm accuracy; ±6° development test range.*)
- [5] E. Singh, “Design and Testing of a Low-Cost 3D-Printed Servo Gimbal for Thrust Vector Control in Model Rockets,” arXiv:2509.00061, August 2025. <https://arxiv.org/abs/2509.00061> (*±5° deflection, 44.5 ms mean response with SG90 servos, 30 N thrust load, PLA→ABS material change and its rationale.*)
- [6] Oriental Motor, “The Choice Between Servo Motors and Stepper Motors.” <https://blog.orientalmotor.com/the-choice-between-servo-motors-and-stepper-motors> (*Load-to-rotor inertia ratios: ~10× open-loop stepper, ~30× closed-loop stepper, ~100× servo; open-loop step-loss failure mode.*)

- [7] StepperOnline (OMC), NEMA 17 hybrid stepper motor catalogue and datasheets. <https://www.omc-stepperonline.com/nema-17-stepper-motor> (*Holding torque 16–79 N cm across the 42 mm frame; rotor inertia, rated current, and physical dimensions used in the actuator catalogue.*)
- [8] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed., Wiley, 2017. (*LOX/RP-1 chamber gas properties, Table 5-5, as used by `raosim/propellants.py`; gimbal TVC as the dominant large-vehicle method.*)
- [9] NASA SP-125, *Design of Liquid Propellant Rocket Engines*. (*Characteristic length $L^* = 40\text{--}50$ in for LOX/RP-1, Table 4-1; stainless steel named as a common thrust-chamber wall metal, p. 105; station-wise liner stress screens, eqs. 4-28/29/31/32 — all as cited inside LREKit’s own source.*)
- [10] NASA SP-8120, *Liquid Rocket Engine Nozzles*. (*Throat-approach radius-ratio bounds and the Rao/TOP chart domain enforced by `raosim/throat_geometry.py`.*)
- [11] I. Shahid, *LREKit / RaoRocketSim*. Modules cited directly: `raosim/materials.py` (316L catalogue entry), `raosim/run_nozzle.py` (traditional workflow), `raosim/mdo/` (`engine.py`, `nlp.py`, `snapshot.py`, `postprocess.py`), `docs/MDO_GUIDE.md`, `scripts/tvc_sizing_study.py` (added by this study).